Marmara University
Engineering Faculty, Electrical and Electronics Engineering
EE4077 Fundamentals of Machine Learning, Midterm Exam, Spring, 2025

1. A certain experiment produces the following data

x	y
1	2.5
2	4.3
3	5.5
4	6.1
5	6.1

where x is the observed variable, y is the target variable. We will use **nonlinear regression** to fit a model of the form:

$$\hat{y} = w_1 x + w_2 x^2$$

to the observed data. The error criterion is l_2-norm.

(a) [6 pt] Give the observation matrix X, target vector y, and the unknown parameter vector w.
(b) [6 pt] Find the associated optimum model $w^* = \left(X^T X\right)^{-1} X^T y$.
(c) [4 pt] Is the model unique?
(d) [4 pt] Do we need to apply **ridge regression** for the observed data?

2. Suppose you are given the following binary classification task: predict the target $Y \in \{0,1\}$ given two real valued features $X_1 \in \mathbb{R}$ and $X_2 \in \mathbb{R}$. After some training, you learn the following linear decision rule

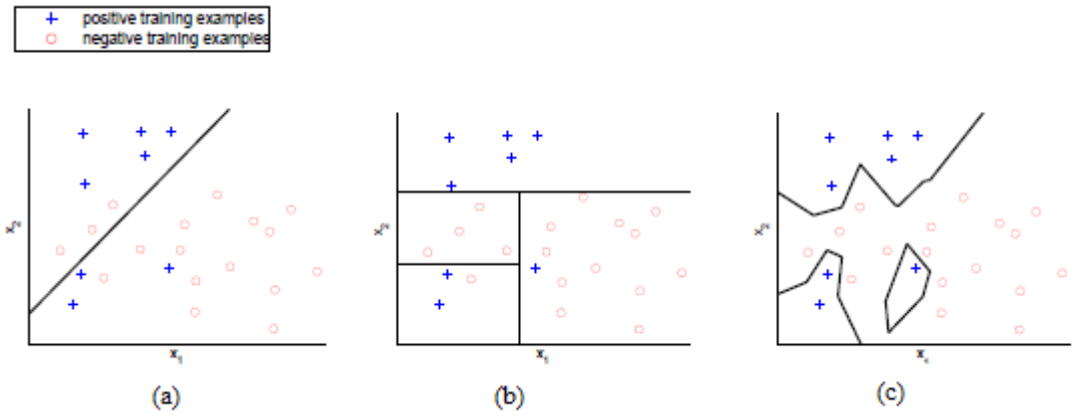
$$\hat{Y} = \begin{cases} 1, & w_1 X_1 + w_2 X_2 + w_0 \geq 0 \\ 0, & \textit{otherwise} \end{cases}$$

where $w_1 = 3$, $w_2 = 5$, $w_0 = -15$.

(a) [10 pt] Plot the decision boundary and label the region where we would predict $Y = 1$ and $Y = 0$. In the plot, the horizontal and vertical axes represent X_1 and X_2, respectively.
(b) [10 pt] Suppose that we learned the weights using logistic regression. What would be our prediction for Y if $X_1 = 3$, $X_2 = 9$? Use the sigmoid function given below to compute

$$P[Y = 1] = \frac{1}{1 + e^{-(w_1 X_1 + w_2 X_2 + w_0)}}$$

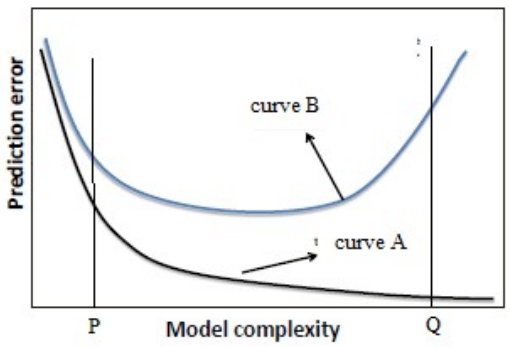
3. The following figures depict decision boundaries of classifiers obtained from three learning algorithms: decision trees, logistic regression, and nearest neighbor classification.
(a) [7 pt] In (a), what is the classifier, how many mistakes do it make on the training data?
(b) [7 pt] In (b), what is the classifier, how many mistakes do it make on the training data?
(c) [6 pt] In (c), what is the classifier, how many mistakes do it make on the training data?



4. In this problem, you will see how to learn a decision tree. Consider the following data set with two binary features X_1, X_2, and three classes $Y \in \{1,2,3\}$.
   (a) [6 pt] Which feature, X_1 or X_2, provides more information gain? Shortly explain your answer without calculating the corresponding entropies.
   (b) [10 pt] Which feature is used for the first split? Draw the decision tree resulting from using this split alone. Make sure to label the split attribute, which branch is which, and what the predicted label is in each leaf.
   (c) [4 pt] How would this tree classify an example with $X_1 = 0$ and $X_2 = 1$?

X_1	X_2	Y
1	1	1
1	1	1
1	1	2
1	0	3
0	0	2
0	0	3

5. Answer the following questions for the given plot. Answers without justification will get no credit.
   (a) [4 pt] **True/False.** Curve A is more likely to be the validation error.
   (b) [4 pt] **True/False.** Curve B is more likely to be the training error.
   (c) [4 pt] **True/False.** At complexity P, the model has high bias and low variance
   (d) [4 pt] **True/False.** At complexity Q, the model has low bias and high variance
   (e) [4 pt] **True/False.** At complexity P, the model overfits while at complexity Q it underfits.



Directions:
1. Closed book, closed notes separating
2. The exam is 75 minutes.

3. Justify your answers.